Rajalakshmi Engineering College

FUNDAMENTALS OF DATA SCIENCE LAB

EXPERIMENT 11

Understand pandas library use cases in Data Science

Experiment 11: Understand pandas library use cases in Data Science

Aim

To demonstrate data frame operations using pandas.

Source Code

```
import numpy as np
import pandas as pd
list=[[1,'Smith',50000],[2,'Jones',60000]]
df=pd.DataFrame(list)
df
df.columns=['Empd','Name','Salary']
df
df.info()
df=pd.read_csv("/content/50_Startups.csv")
df.info()
df.head()
df.tail()
import numpy as np
import pandas as pd
df=pd.read_csv("/content/employee.csv")
df.head()
df.tail()
df.info()
df.salary
salary
type(df.salary)
pandas.core.series.Series
def __init__(data=None, index=None, dtype: Dtype | None=None, name=None, copy: bool | None=None,
fastpath: bool=False) -> None
df.salary.mean()
df.salary.median()
df.salary.mode()
salary
df.salary.var()
df.salary.std()
df.describe()
df.describe(include='all')
empCol=df.columns
empCol
Index(['emp id', 'name ', 'salary'], dtype='object')
emparray=df.values
emparray
employee_DF=pd.DataFrame(emparray,columns=empCol)
employee_DF
```

Output - Experiment 11

0 1 2
0 1 Smith 50000
1 2 Jones 60000
Empd Name Salary
0 1 Smith 50000
1 2 Jones 60000
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 2 entries, 0 to 1
Data columns (total 3 columns):
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 50 entries, 0 to 49
Data columns (total 5 columns):
R&D Spend Administration Marketing Spend State Profit
0 165349.20 136897.80 471784.10 New York 192261.83
1 162597.70 151377.59 443898.53 California 191792.06
2 153441.51 101145.55 407934.54 Florida 191050.39
3 144372.41 118671.85 383199.62 New York 182901.99
4 14210734 9139177 36616842 Florida 16618794
R&D Spend Administration Marketing Spend State Profit
45 1000.23 124153.04 1903.93 New York 64926.08
46 1315.46 115816.21 297114.46 Florida 49490.75
47 0.00 135426.92 0.00 California 42559.73
48 542.05 51743.15 0.00 New York 35673.41
49 000 11698380 4517306 California 1468140
emp id name salary
0 1 SREE VARSSINI K S 5000
1 2 SREEMATHI B 6000
2 3 SREYA G 7000
3 4 SREYASKARI MULLAPUDI 5000
4 5 SRIAKASHUG 8000
emp id name salary
2 3 SREYA G 7000
3 4 SREYASKARI MULLAPUDI 5000
4 5 SRI AKASH U G 8000
5 6 SRI HARSHAVARDHANAN R 3000
6 7 SRIHARSHAVARDHANANR 6000
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 7 entries, 0 to 6
Data columns (total 3 columns):
0 5000
1 6000
2 7000
3 5000
4 8000
5 3000
6 6000
One-dimensional ndarray with axis labels (including time series).
Labels need not be unique but must be a hashable type. The object
supports both integer- and label-based indexing and provides a host of
methods for performing operations involving the index. Statistical
5714.285714285715

Output - Experiment 11 (continued)

```
6000.0
0 5000
1 6000
2571428.5714285714
1603.5674514745463
emp id salary
count 7.000000 7.000000
mean 4.000000 5714.285714
std 2.160247 1603.567451
min 1.000000 3000.000000
25% 2.500000 5000.000000
50% 4.000000 6000.000000
75% 5.500000 6500.000000
max 7000000 8000000000
emp id name salary
count 7.000000 7 7.000000
unique NaN 6 NaN
top NaN SRI HARSHAVARDHANAN R NaN
freq NaN 2 NaN
mean 4.000000 NaN 5714.285714
std 2.160247 NaN 1603.567451
min 1.000000 NaN 3000.000000
25% 2.500000 NaN 5000.000000
50% 4.000000 NaN 6000.000000
75% 5.500000 NaN 6500.000000
max 7000000 NaN 8000000000
array([[1, 'SREE VARSSINI K S', 5000],
[2, 'SREEMATHI B', 6000],
[3, 'SREYA G', 7000],
[4, 'SREYASKARI MULLAPUDI', 5000],
[5, 'SRI AKASH U G', 8000],
[6, 'SRI HARSHAVARDHANAN R', 3000],
[7, 'SRI HARSHAVARDHANAN R', 6000]], dtype=object)
emp id name salary
0 1 SREE VARSSINI K S 5000
1 2 SREEMATHI B 6000
2 3 SREYA G 7000
3 4 SREYASKARI MULLAPUDI 5000
4 5 SRI AKASH U G 8000
5 6 SRI HARSHAVARDHANAN R 3000
6 7 SRIHARSHAVARDHANANR 6000
```

Result - Experiment 11

Pandas library use cases were demonstrated using data frames.